MATH7005 2026/2027
Assignment and Project Specification

5 October 2026

1 Key information

- **Initial release date:** 5 October 2026.
- **Assignment deadline:** 26 October 2026.
- **Project submission deadline:** the last day of class.
- This is an **individual assessment**. Each student must develop and submit their own work.
- The assignment accounts for **10%** and the project accounts for **90%** of the subject's final mark.
- The principal project assessment, including the live demonstration and question-and-answer session, will take place in late November. Scheduling and further assessment details will be announced in late October.

2 Introduction

In MATH7005, you have learned a range of essential data analytics and machine learning techniques, together with substantial Python programming. In particular, you have developed practical experience with NumPy and pandas for data manipulation, analysis, and computation. These skills are highly useful in both academic and business settings. The assignment and project give you an opportunity to apply them to a real-world problem by proposing and developing a data product.

Entrepreneurial experience is highly valued in today's job market. You are encouraged to consider how your work could become a viable venture.

The assessment is designed to strengthen your data analytics, machine learning, product-design, coding, communication, and critical-thinking skills. Each student must design and build at least part of a data analytics or machine learning tool centred on the annual theme.

3 Theme

For the 2026–2027 assessment, the theme is:

> *A Python-powered data product using Hong Kong open data to improve life, services, or economic opportunities in Hong Kong.*

The theme is deliberately broad enough to support diverse ideas while retaining a clear focus on Hong Kong and the practical use of local open data. Projects may address areas such as transport, public services, health, the environment, housing, education, culture, tourism, commerce, or intellectual property. The proposed product must address a clearly defined problem and provide value to a specific group of users.

4 Assignment: video pitch (10%)

Each student must submit a three-minute video pitch consisting of voice-over slides. The pitch should establish the context for the data product that the student proposes to develop for the project.

The pitch should concisely explain:

- the problem or opportunity being addressed;
- the intended users and their needs;
- the proposed data product and its value;
- the relevant data that may be used; and
- the intended social, economic, or practical impact.

This lightly weighted task is intended to keep students on track and to support the early development of a feasible project. The deadline is **26 October 2026**. Submission instructions will be provided on Moodle.

5 Project (90%)

5.1 Problem setting and required work

Imagine that you are a startup founder building an innovative, publicly usable data-mining tool. The tool may take the form of a website, mobile application, desktop application, smart device, or another suitable product. Your work should include, but is not limited to, the following:

1. Design a data product for the theme. Address fundamental questions such as:
    - Who are the intended users?
    - Is the product goal realistic?
    - What makes the product unique?
    - What impact could the product have on society?
2. Select suitable datasets. Every project **must use one or more datasets from the Hong Kong Government's open-data portal, https://data.gov.hk/en/, as a main part of its data and analysis**. Supplementary data may also be obtained from sources such as:
    - datasets from Kaggle, such as https://www.kaggle.com/;
    - intellectual property datasets; and
    - other sources relevant to the theme.
3. If the available data are insufficient, propose additional "good-to-have" datasets. Specify the key attributes or fields, explain why they are important, and identify who may be able to provide the data. You may use Python to generate realistic synthetic data where appropriate, but you must clearly label and justify such data.
4. Propose and build at least a partial product prototype. A fully functional product is not required, but the prototype must contain working features.
5. Apply relevant modelling and programming techniques to the downloaded and/or generated data, and explain how each technique advances the product goals. All submitted code must be written in Python.
6. **For students who wish to achieve an outstanding mark:** Research the machine learning literature and select an original research paper published in or after 2022 that has more than 2,022 citations. Survey and review papers do not count. Discuss how the selected paper's techniques could be applied to your data product.

7. Use suitable data-mining, data-science, machine-learning, or AI methods from this subject or beyond it. However, do not base the project primarily on techniques covered in other ORBS subjects. For example, a project primarily based on ARIMA and the elbow method for PCA will receive a very low score.

8. Propose future steps needed to bring the data product to life.

5.2 Project deliverables and assessment

1. **Final report and code**

   Submit a report documenting the steps taken to achieve your goals. Extensive background material is unnecessary unless it primarily reflects your own contribution.

   - There is no word limit, but the report should comprehensively record what you did.
   - Include the complete code as an attachment or in the format specified on Moodle.
   - You may describe failed attempts when they demonstrate effort, reflection, or learning.
   - No resubmission is allowed after the deadline.
   - The report and associated materials must be submitted on Moodle by the **last day of class**.

2. **Live product demonstration**

   Each student will give a **10-minute live demonstration** of their finished data product. **Slides are not permitted.** The demonstration should focus on the working product, the underlying data analysis, and the Python code.

3. **Question-and-answer session**

   The demonstration will be followed by an approximately **20-minute AI-assisted question-and-answer session** with the assessor. As part of the assessment process, AI will collaborate with the human assessor to analyse each student's submitted report and formulate questions based on that student's own submitted content. The questions will therefore be individualised. Each student must be able to explain and defend all parts of their own work, including the design choices, modelling, implementation, code, results, limitations, and use of AI.

5.3 Use of AI

AI assistance is allowed for the project, provided that it is clearly declared. The final report must identify the AI tools used and describe how they were used. Students remain responsible for the correctness, originality, and integrity of all submitted work and must be able to explain every part of their code and analysis during the live assessment, regardless of whether the code was written by the student or generated with the assistance of AI.

6 Grading

The project will be graded broadly on the following aspects:

1. **Understanding of subject knowledge.** Demonstrate a broad set of relevant skills by applying multiple data-mining tasks to the proposed data product. Do not limit the work to only one or two methods.

2. **Understanding of Python coding.** Demonstrate a thorough understanding of all submitted Python code, including code written by the student and code generated or modified with AI assistance.

   - Code should be well documented and easy to understand.
   - Main Python functions should meaningfully implement the algorithms. One-line calls to scikit-learn functions will not be accepted as the primary implementation.
   - Code must be clear enough to explain during the live demonstration and question-and-answer session.

3. **Uniqueness and innovation.** Explain why the data product is distinctive, innovative, and useful. The "productness" of the work is important: do not submit something that is merely a report and cannot be used by its intended audience.

4. **Report quality.** Present the problem, data, methods, product, findings, limitations, AI use, and future work clearly and rigorously.

5. **Live demonstration and responses.** Demonstrate a functioning product and answer questions accurately, critically, and independently.

6. **Originality and academic integrity.** Every student is expected to submit a distinct solution. Discussion with classmates is understandable, but sharing work or submitting substantially similar products is not acceptable and may result in reduced marks or further action under the University's academic integrity procedures.